Unit 9, Lesson 1: Finding the Area of Two-Dimensional Figures

Benchmark

MA.912.GR.4.4 Solve mathematical and real-world problems involving the area of two-dimensional figures.

Learning Target

- [ ] I can solve mathematical and real-world problems involving the area of two-dimensional figures.

9.1.1 Warm-Up: Michael Jordan

Basketball legend Michael Jordan is often quoted as saying, “I’ve failed over and over and over again in my life and that is why I succeed.”

1. What is a skill, activity, or subject that you have failed at?

2. Explain Jordan’s quote in your own words.

3. How can you turn a failure into a success story?

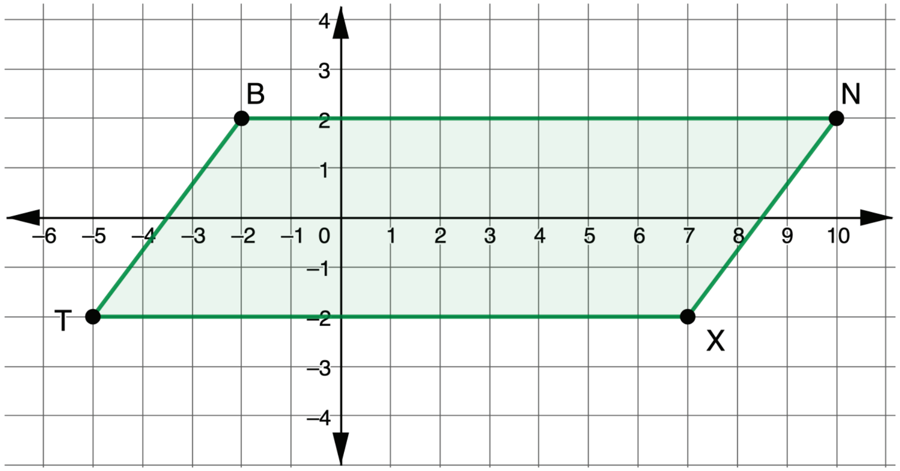
9.1.2 Collaboration: Finding Area

1. Complete the table.

Figure	Area Formula
Rectangle	
Triangle	

2. How is the area of a triangle related to the area of a rectangle?

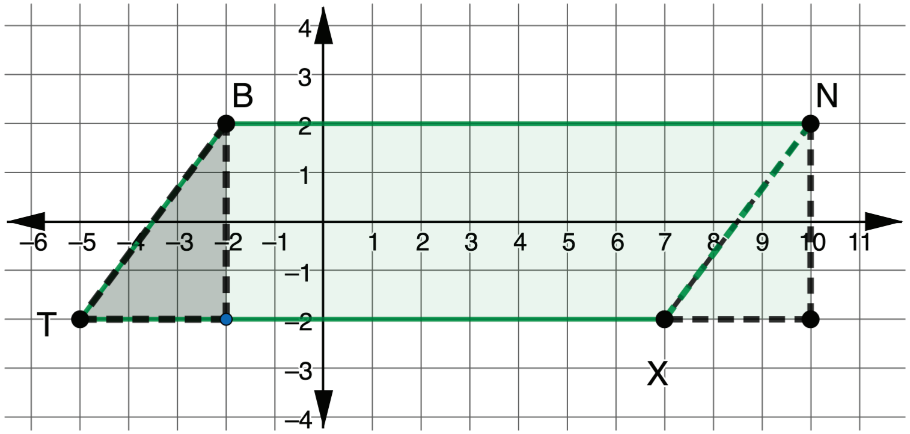
3. Parallelogram $BNXT$ is shown on the coordinate grid.



Tyrese could not remember how to find the area of a parallelogram so he divided the parallelogram into 2 triangles and a rectangle. Work with your partner to use Tyrese's method to find the area of $BNXT$.

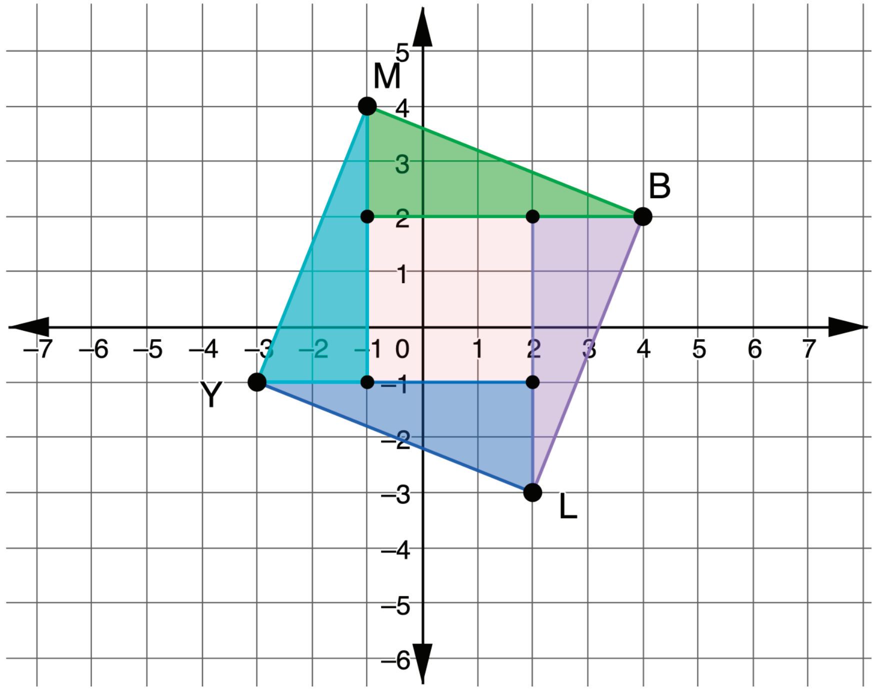
Shape	Area
Triangle 1	
Triangle 2	
Rectangle	
Total Area of Parallelogram	

4. Sarianna rearranged the parallelogram to create a rectangle.

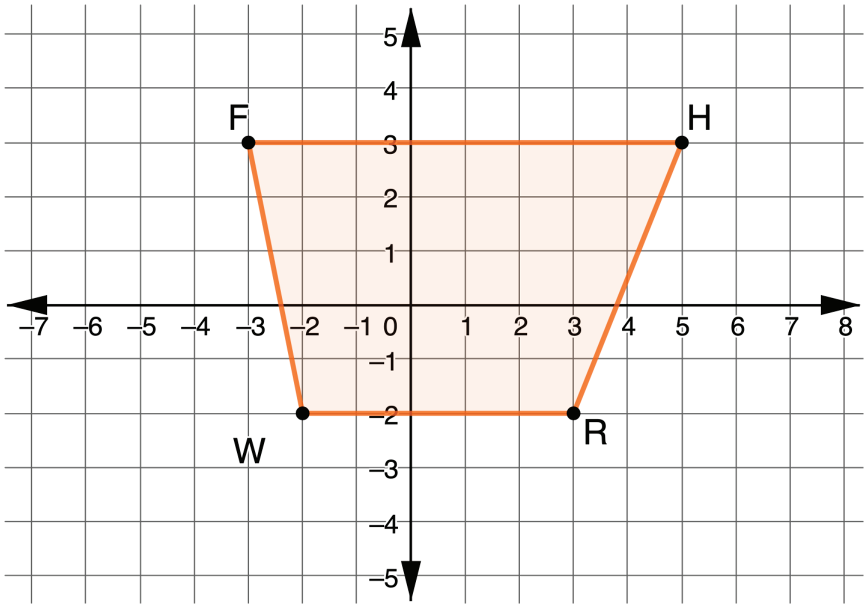


Use Sarianna's rectangle to find the area of $BNXT$.

5. Quadrilateral $MBLY$ is decomposed as shown on the coordinate grid. Find the area of $MBLY$ using the decomposition shown.



6. $FHRW$ is shown on the coordinate grid.



a. Divide the quadrilateral into two triangles and a rectangle. Then, find the area of each.

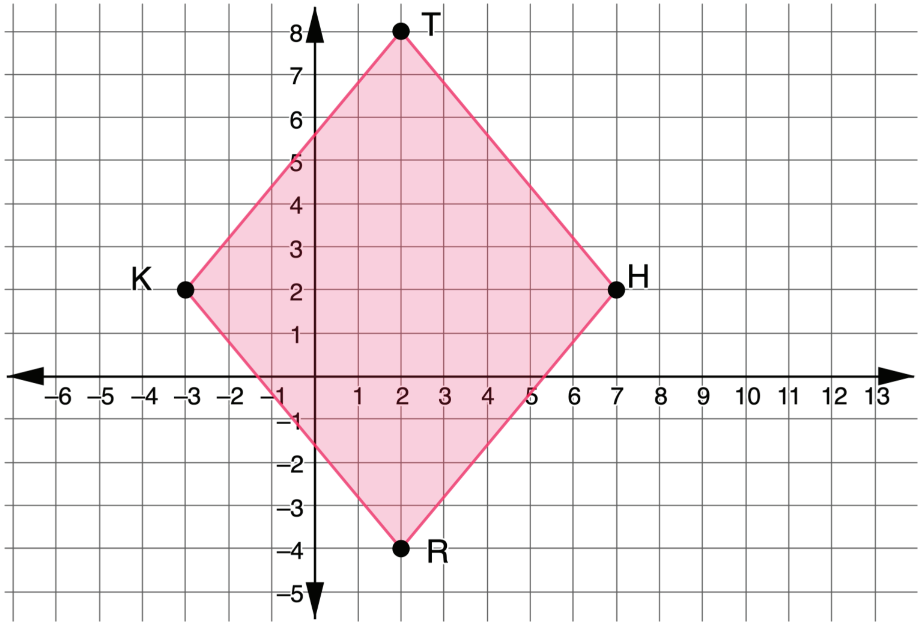
Shape	Area
Triangle 1	
Triangle 2	
Rectangle	

b. What is the area of $FHRW$?

c. What type of quadrilateral is $FHRW$?

7. Ava found a formula for the area of a trapezoid. The formula is $A = \frac{1}{2}h(b_1 + b_2)$, where h is the height and b_1 and b_2 are the bases. Use the formula Ava found to find the area of $FHRW$.

8. $THRK$ is shown on the coordinate grid.



a. What type of quadrilateral is $THRK$?

b. Determine the area of $THRK$. Show your work or explain your thinking.

c. Complete the table.

Line Segment	Length
$\overline{TR}$	
$\overline{KH}$	

The area of a rhombus can be found using the formula $A = \frac{1}{2}pq$, where p and q are the lengths of the diagonals.

d. Use the formula to find the area of $THRK$.

Formulas for area can easily be forgotten, so it is always helpful to be able to break a figure down into triangles and rectangles in order to find the area. The exception is finding the area of a circle or a part of a circle.

9. What is the area of a circle?

10. The drama program at school is building a new semicircular stage. The longest distance across the stage is 30 feet. What is the area of the new stage? Round to the nearest thousandth.

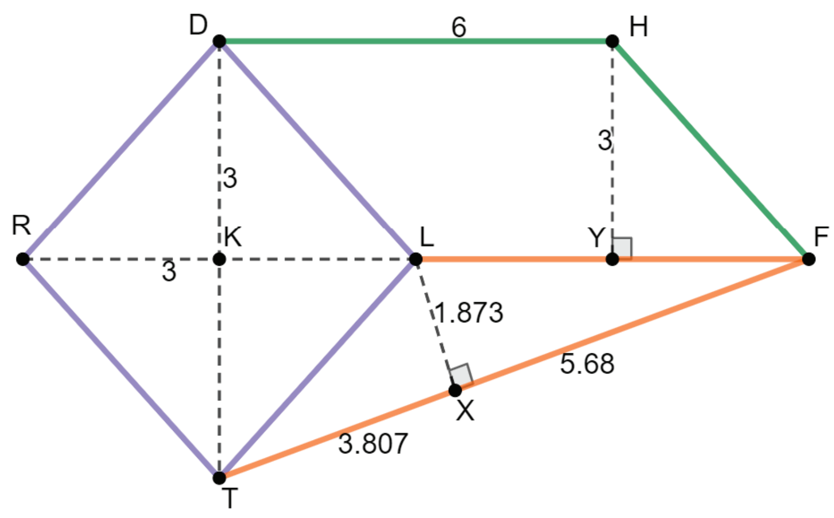
9.1.3 Guided Practice: Using the Formulas

1. The radius of a circle is 7 feet (ft). Find the area, in square feet, of the circle. Leave the answer in π form.

2. The base of the parallelogram is 22 inches (in.). The area of the parallelogram is 132 square inches. Find the height, in inches, of the parallelogram.

3. The diagonals of a rhombus are 40 centimeters (cm) and 18 centimeters. Find the area, in square centimeters, of the rhombus.

4. A composite shape of a rhombus, a parallelogram, and a triangle is shown.

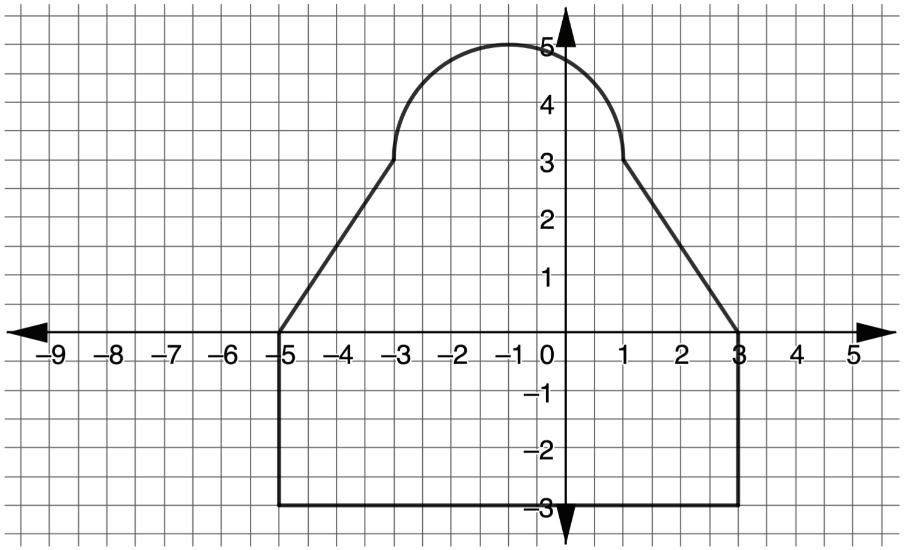


Complete the table. Round to the nearest thousandth.

Shape	Area
Rhombus	
Parallelogram	
Triangle	
Composite Area	

9.1.4 Your Turn!

1. The design of a room is shown, where 1 unit $= 5$ feet.



Determine the area, in square feet, of the room. Round to the nearest hundredth.

9.1.5 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each.

I need help. I got it. I could help someone.

Finding the area of a triangle	
Finding the area of a rectangle	
Finding the area of a trapezoid	
Finding the area of a square	
Finding the area of a rhombus	
Finding the area of a parallelogram	
Finding the area of a circle	
Finding the area of a composite figure	